Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/20/2004
PI Name:	Packard, Richard E. Ph.D.		
Project Title:	Development of High Sensitivity Sensors of Absolute Rotation using Quantum Phase Coherence In superfluid Helium		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	04/01/2003	**End Date:**	09/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We will enhance the sensitivity of a superfluid gyroscope based on a dc-SQUID device developed in an earlier grant. We hope to determine the practical limitation of this technology. 1. Determine the current-phase characteristic of He3 weak links at elevated ambient pressure in order to find the optimal operating point for a superfluid gyroscope. 2. Enhance the sensitivity of the superfluid gyroscope by a factor of 10-100 over pre-existing devices. The superfluid gyroscope can be the most sensitive sensor of rotation extant. Only a prototype device has been demonstrated previously. If the superfluid gyroscope can reach its ultimate theoretical sensitivity it will enhance measurements of the Earth's rotation vector, which ultimately provides a limitation on the GPS navigation system.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the grant's second year we have constructed our test cell for experiments on He3 weak links. During the testing of the device in superfluid He4 we discovered the phenomenon of Josephson oscillations in He4 near 2K. This is a very major breakthrough for the project. It suggests that a sensitive superfluid gyroscope (an analoge of a dc SQUID) can be developed using He4 rather than He3. The He4 device operates at a temperaure 2000 times warmer than the He3 technology. This may result in an enormous simplification in the ultimate gyroscope. Operating at these higher temperatures makes it more likely that a precision superfluid gyroscope would be attractive to users who have little expertise in low temperture technology. We are presently beginning our investigation of the new He4 quantum whistle. We have performed experiments on the He4 quantum whistle. We have determined that the frequency is given precisely by the Josephson frequency formula. We have partially completed analysis of these experiments.
Bibliography Type:	Description: (Last Updated: 09/12/2008)
Articles in Peer-reviewed Journals	Penanen,K;Hoffman, J;Davis, J.C.; Packard, R.E. "Effect of 3He on third sound attenuation in thick 4He Films" J. of Low Temp. Phys., 134, 1069 (2004) , Mar-2004
Articles in Peer-reviewed Journals	Penanen, K.;Hoffman, J.; Davis, J.C.;Packard, R.E. "104. Measurement of the Attenuation of Third sound: Evidence of Trapped Vorticity in Thick Films of Superfluid 4He" J. of Low Temp. Phys, 135, 177 (2004) , May-2004
Articles in Peer-reviewed Journals	Packard,R.E. "Phase slips and Josephson weak links in superfluid helium" J. of Low Temp. Phys., 135, 471 2004 , Jan-2004
Articles in Peer-reviewed Journals	R. Ishiguro, O. Ishikawa, M. Yamashita, Y. Sasaki, K. Fukuda, M. Kubota, H. Ishimoto, R.E. Packard, T. Tagaki T. Ohmi, T. Mizusaki "Vortex formation and annihilation in three textures of rotating superfluid 3He-A" Phys. Rev. Letters 93, 125301, (2004) , Jan-2004
Articles in Peer-reviewed Journals	Hoskinson,E. , Packard, R.E. "Quantum Whistles in Superfluid 4He" Nature , Feb-2005